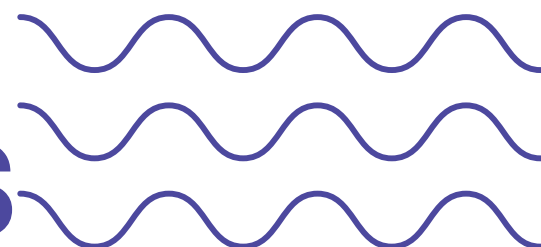




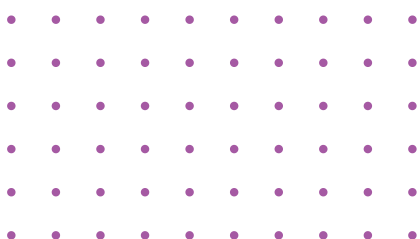
PROJECT 4

GIGA GIRLS



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THE PROJECT'S GOAL:

In this project, we designed and implemented a comparator circuit that receives two 3-bit inputs (**A2 A1 A0 and B2 B1 B0**) and gives the following output cases:

Case 1: A = B

1- $A_2=B_2$ AND $A_1=B_1$ AND $A_0=B_0$

2- $X_i = (A_i' \cdot B_i' + A_i \cdot B_i) = A_i(.)B_i$

$X_0 = A_0(.)B_0$, $X_1 = A_1(.)B_1$, $X_2 = A_2(.)B_2$

$$\underline{X} = (\underline{X_0} \cdot \underline{X_1} \cdot \underline{X_2})$$

Case 2: A > B

1- $(A_2=1$ AND $B_2=0)$

2- OR $(A_2=B_2)$ AND $(A_1=1$ AND $B_1=0)$

3- OR $(A_2=B_2$ AND $A_1=B_1)$ AND $(A_0=1$ AND $B_0=0)$

$$\underline{Y} = (\underline{A_2} \cdot \underline{B_2}') + (\underline{X_2} \cdot \underline{A_1} \cdot \underline{B_1}') + (\underline{X_1} \cdot \underline{X_2} \cdot \underline{A_0} \cdot \underline{B_0}')$$

Case 3: A < B

1- $(A_2=0$ AND $B_2=1)$

2- OR $(A_2=B_2)$ AND $(A_1=0$ AND $B_1=1)$

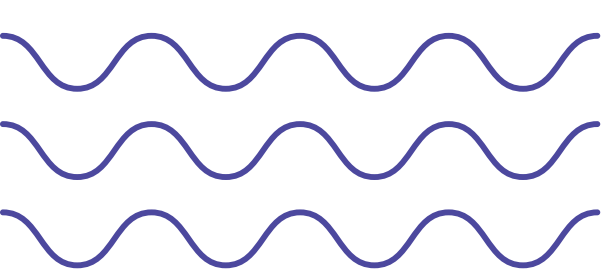
3- OR $(A_2=B_2$ AND $A_1=B_1)$ AND $(A_0=0$ AND $B_0=1)$

$$\underline{Z} = (\underline{A_2}' \cdot \underline{B_2}) + (\underline{X_2} \cdot \underline{A_1}' \cdot \underline{B_1}) + (\underline{X_1} \cdot \underline{X_2} \cdot \underline{A_0}' \cdot \underline{B_0})$$

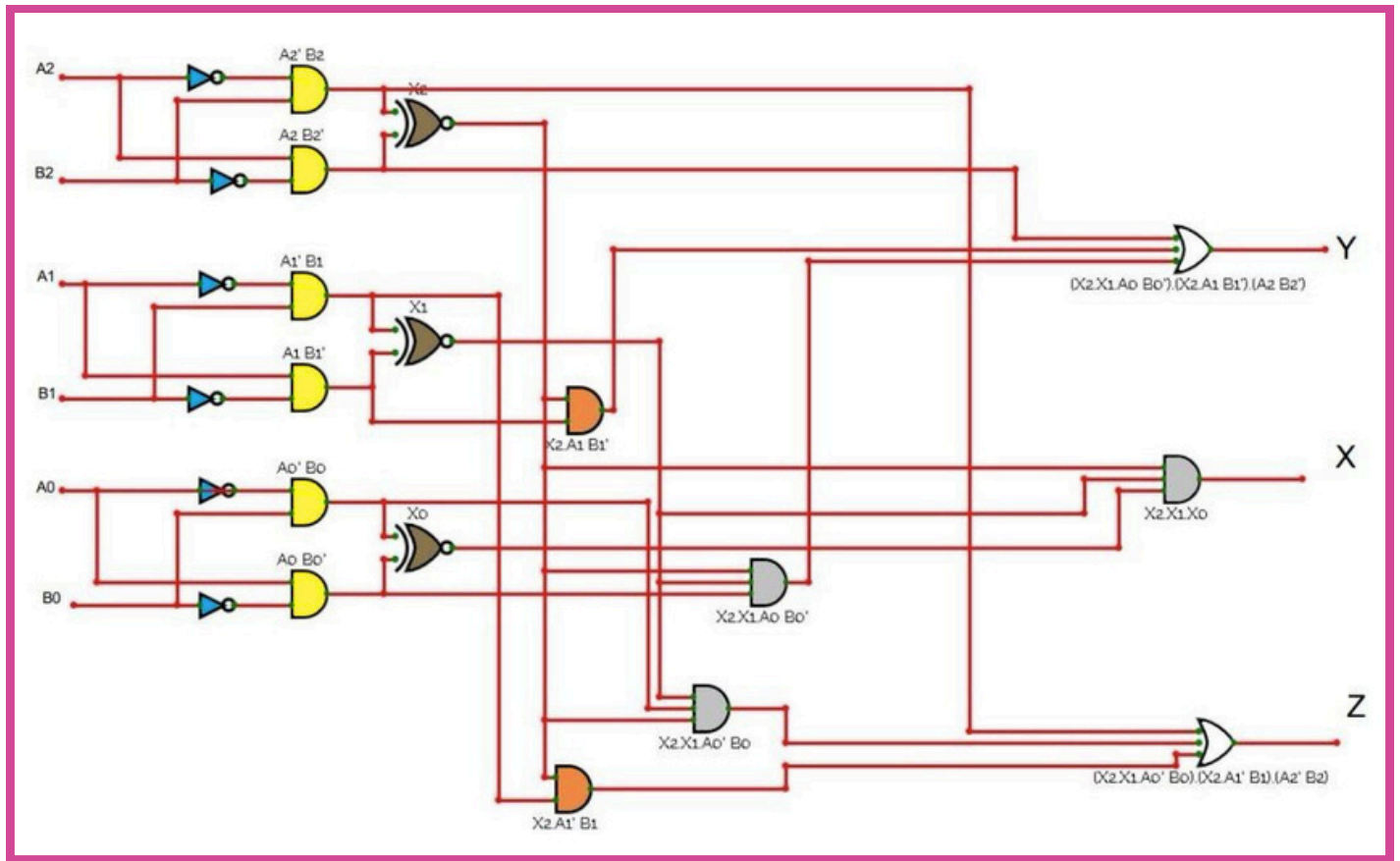


COMPONENTS OF THE CIRCUIT

Item	Funtion
DIP Switch	<i>Turns the circuit on and off</i>
NOT Gate	<i>Gets the complement of the inputs</i>
2 AND Gates 2 Inputs	<i>Allows the output only if both inputs = 1</i>
XNOR 2 Inputs	<i>Allows the output when both inputs are the same</i>
AND Gate 3 Inputs	<i>Allows the output only if all 3 inputs = 1</i>
OR Gate 3 Inputs	<i>Allows the output if at least one input = 1</i>
Battery 9V / Power Supply	<i>Provides the circuit with electricity</i>
Resistors 10K / 240 ohm	<i>Decreases the voltage to 5V</i>
2 Bread Boards	<i>Used to implement and design the circuit</i>



VISUALISATION OF THE CIRCUIT



COLORS

Green	<i>Positive Connection</i>
Red	<i>Negative Connection</i>
Purple	<i>Switch Connection</i>
Blue	<i>Switch output - NOT input</i>
Yellow	<i>NOT output - AND input</i>
Brown	<i>AND output - XNOR input</i>
Orange	<i>XNOR output - AND input</i>
Grey	<i>XNOR / AND output - AND input</i>
White	<i>AND output - OR input</i>
Black	<i>X (red LED) - Y (green LED) - Z (yellow LED) output</i>